

Combined Effects of Drugs

- A combination of **two** or **more** drugs can result in an **increase** or a **decrease** in response.
- **Increased response:**
 1. Additive effect
 2. Potentiation (supra-additive)
 3. Synergism

1. Additive Effect

The **combined** effect of **two** or **more** drugs is equal to the **sum** of their individual effect.

Effect of drugs **A + B = Effect of drug A + Effect of drug B**
(1+1=2)

For example, ibuprofen and paracetamol as analgesic.

2. Potentiation (**supra-additive**)

The **enhancement** of action of one drug by another drug that is **inactive** is called **potentiation**.

Effect of drugs **A + B > Effect of drug A + Effect of drug B**
(**1+1 > 2**)

For example, levodopa + carbidopa

Carbidopa **inhibits** the breakdown of levodopa; thus, **enhancing** its effects.

3. Synergism

- When **two** or more drugs are administered simultaneously, their **combined** effect is **greater** than that elicited by **either** drug **alone**.
- **For example,**
sulphamethoxazole + trimethoprim



Decreased response (Drug Antagonism)

- In **antagonism**, the effect of one drug is **decreased** or **abolished** in the presence of **another** drug.

1. Physical antagonism:

The opposing action of the **two** drugs is due to their **physical property**, **e.g.**, **activated charcoal** adsorbs **toxic** substances in poisoning.

2. Chemical antagonism:

The opposing action of the two drugs is due to their **chemical property**,

e.g., antacids are **weak bases** that neutralize **gastric acid**;

Chelating agents are complex **metals** and are useful in **heavy metal** poisoning.

3. Physiological (functional) antagonism:

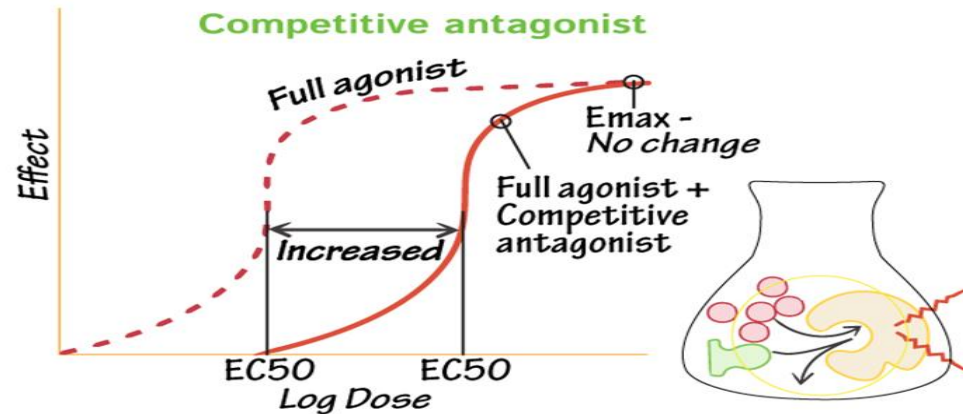
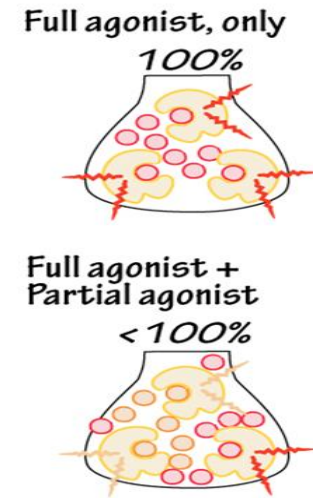
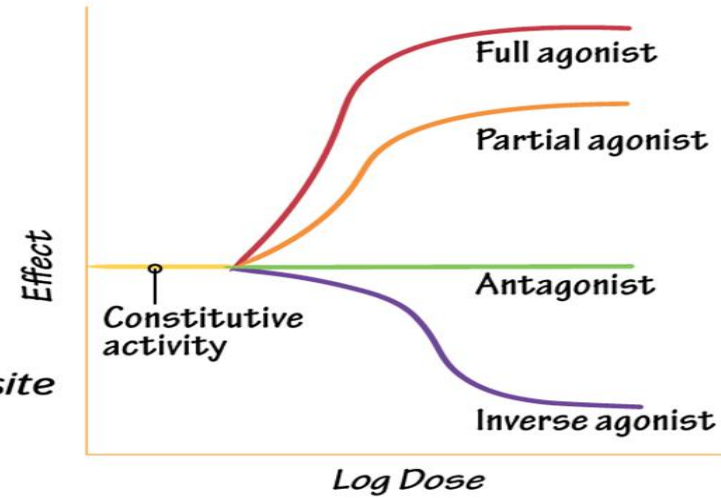
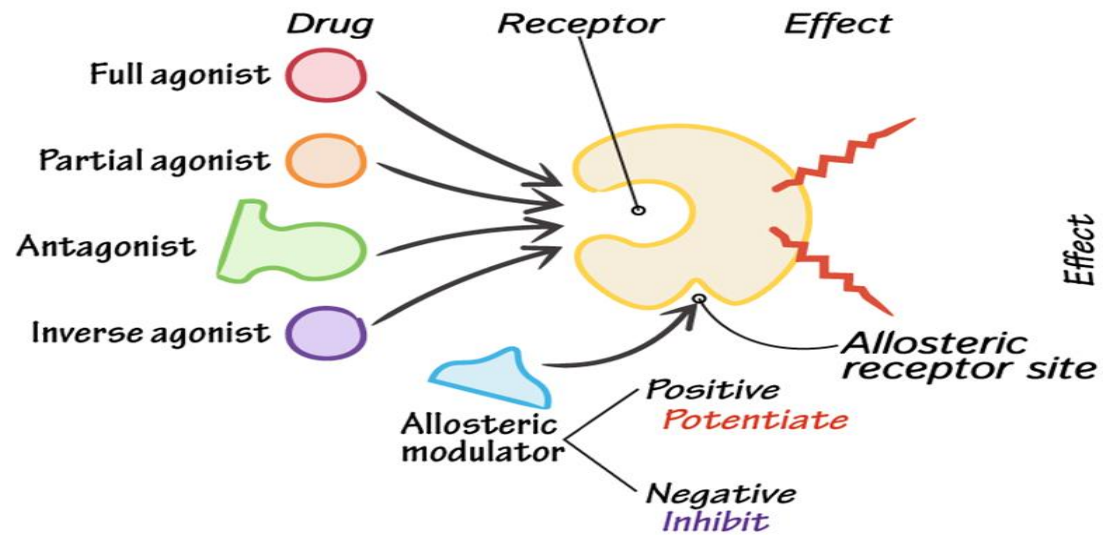
- Here, two drugs act at different **receptors** or by different mechanisms on the same physiological system and produce opposite effects.
 - **For example,**
 - ✓ **insulin** and **glucagon** on blood sugar,
 - ✓ **adrenaline**, and **histamine** on bronchial smooth muscle.
 - ✓ **histamine** produces bronchoconstriction (via histamine receptors), whereas **adrenaline** produces bronchodilatation by acting through adrenergic
- (2) receptors—hence adrenaline helps to reverse bronchospasm in anaphylactic shock.



4. Receptor Antagonism

The **antagonist** binds to the same receptor as the **agonist** and **inhibits** its effects.

It can be **competitive** or **non-competitive**



Increasing the agonist's concentration CAN overcome the antagonist's effect on Emax.



Increasing the agonist's concentration will NOT overcome the antagonist's effect on Emax.

a. *Competitive Antagonism (equilibrium type):*

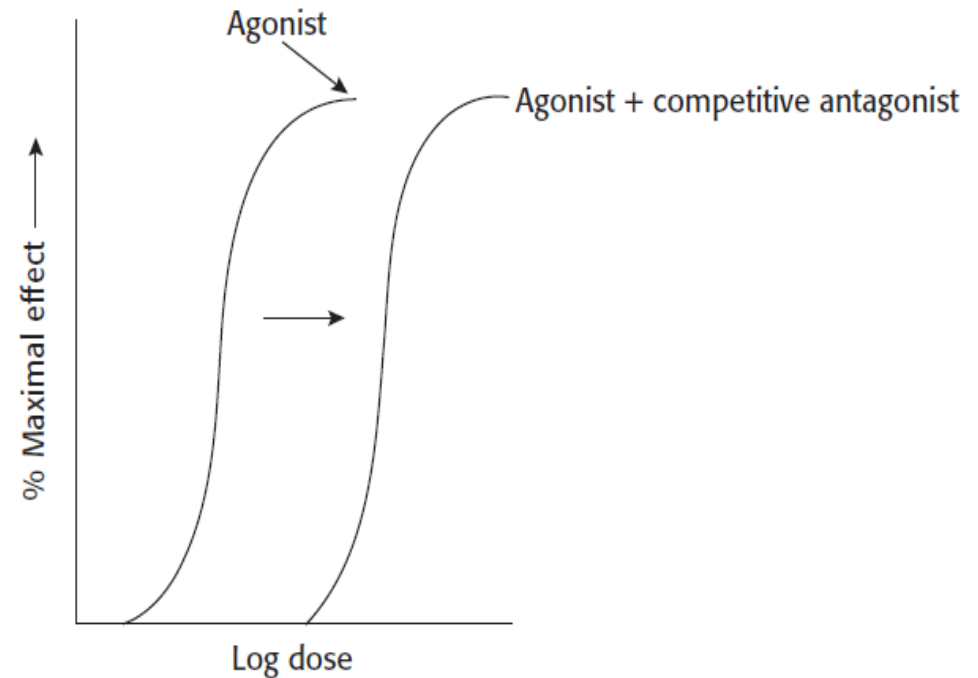
- In **competitive antagonism**, both **agonist** and the antagonist bind reversibly to the same site on the receptor.
- This type of antagonism can be overcome (**reversible**) by increasing the concentration of agonist.
- The log dose-response curve of the agonist shows a rightward parallel shift in the presence of a competitive antagonist

For example,

Acetylcholine	→	Muscarinic receptors	←	Atropine
Morphine	→	Opioid receptors	←	Naloxone
(Agonists)				(Antagonists)

Competitive antagonism

- The log dose-response curve of the agonist shows a rightward parallel shift in the presence of a competitive antagonist





b. *Noncompetitive antagonism:*

- The antagonist binds to a different site on the receptor and prevents the agonist from interacting with the receptor.
 - In this type, the antagonistic effect cannot be overcome by increasing the concentration of the agonist.
 - There is a flattening of the dose-response curve (DRC) in noncompetitive antagonism, e.g. diazepam and bicuculline
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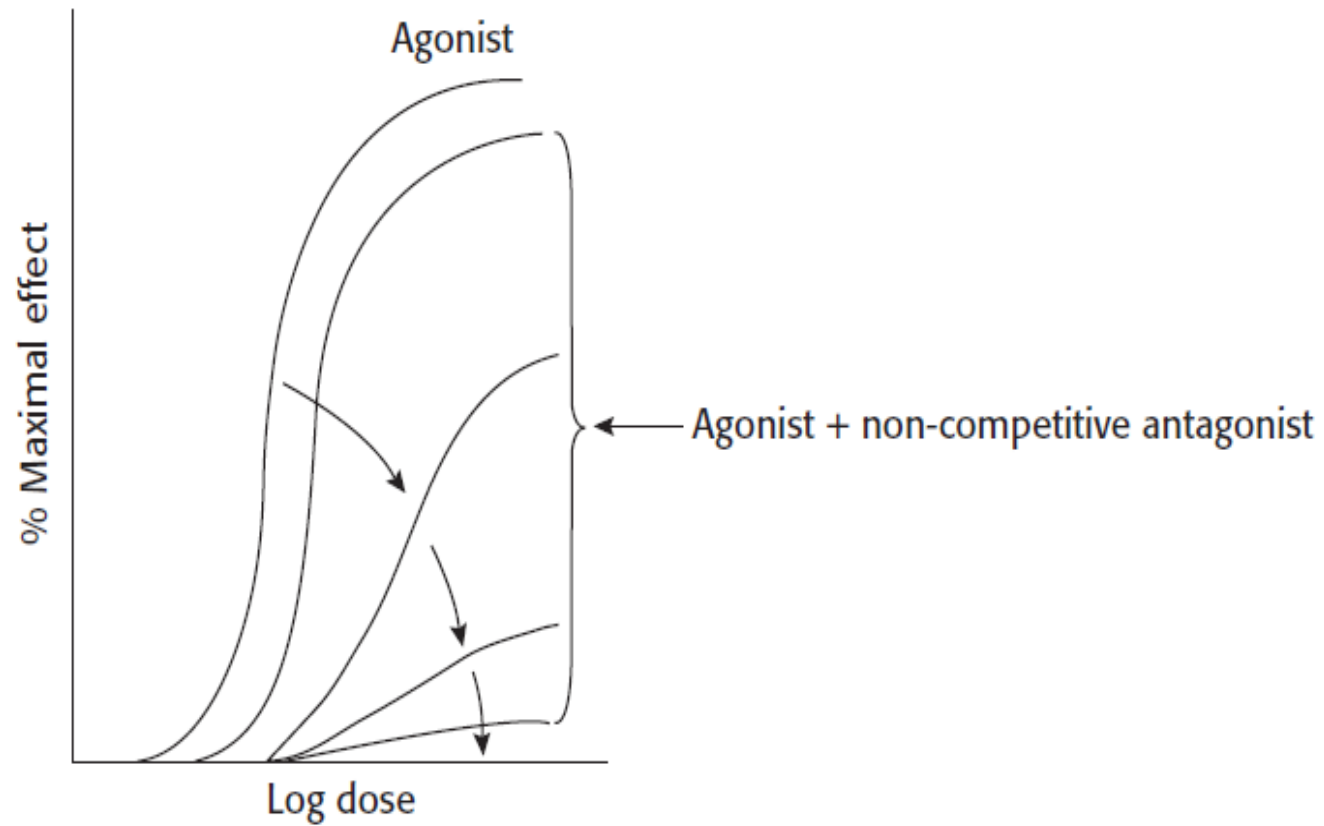


Fig. 1.12 Noncompetitive antagonism.

**Noncompetitive
antagonism.**